



Duration:	60 Minutes	Semester:	Fall 2022
Paper Date:	27/09/2022	Total Marks:	40
Section:	BEE-3A	Weight:	15
Exam Type:	Midterm -1	Page(s):	28

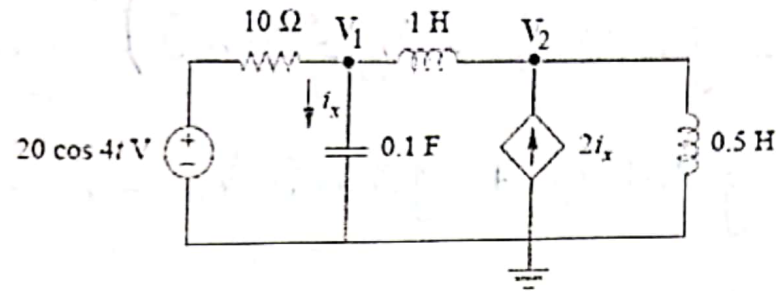
- Carefully understand the question before attempting
- Show all calculations, direct answers are not acceptable

Question # 1

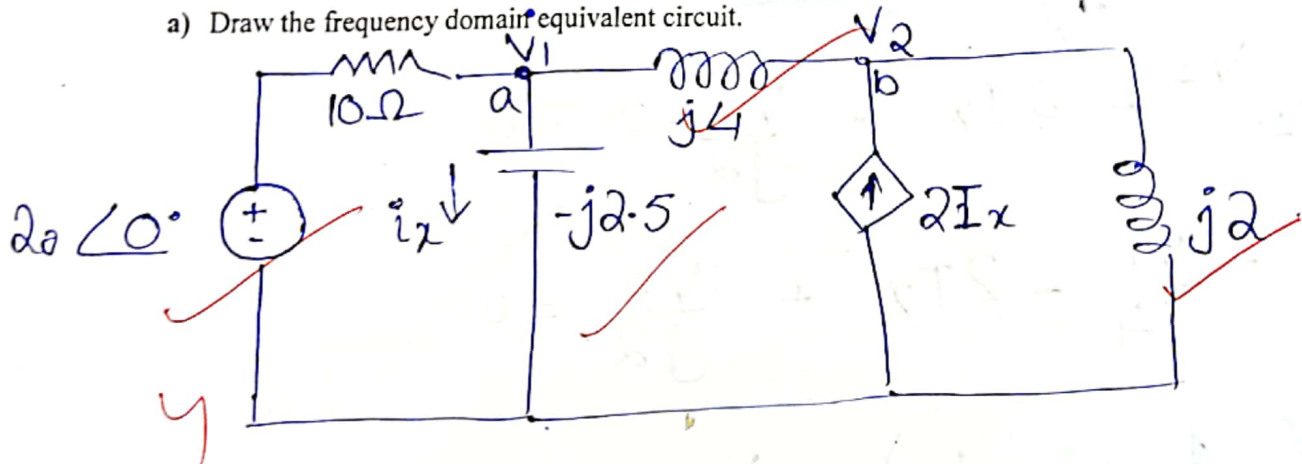
CLO 1

15/ [4 + 10 + 2 + 4 = 20 marks]

Find i_x in the circuit given below using nodal analysis.



a) Draw the frequency domain equivalent circuit.



b) Determine the node voltages, V_1 and V_2 .

Nodal Analysis at node 'a':

$$\frac{V_1 - 20}{10} + \frac{V_1 - 0}{-j2.5} + \frac{V_1 - V_2}{j4} = 0$$

$$\frac{V_1}{10} - \frac{20}{10} + \frac{V_1}{-j2.5} + \frac{V_1}{j4} - \frac{V_2}{j4} = 0$$

$$V_1 \left(\frac{1}{10} - \frac{1}{j2.5} + \frac{1}{j4} \right) - 2 - \frac{V_2}{j4} = 0$$

$$V_1 (0.1 - (-0.4j) + (-0.25j)) - \frac{V_2}{j4} = 2$$

$$V_1 (0.1 + j0.4 - j0.25) - \frac{V_2}{j4} = 2$$

$$V_1 (0.1 + j0.15) - V_2 (j0.25) = 2$$

$$V_1 (0.1 + j0.15) + j0.25 V_2 = 2 \rightarrow (1)$$

Nodal Analysis on node "b" :-

$$\frac{V_2 - V_1}{j4} - 2I_x + \frac{V_2}{j2} = 0$$

$$\frac{V_2}{j4} - \frac{V_1}{j4} - 2I_x + \frac{V_2}{j2} = 0$$

$$V_2 \left(\frac{1}{j4} + \frac{1}{j2} \right) - \frac{V_1}{j4} - 2I_x = 0$$

$$V_2 (-j0.25 - j0.5) - V_1 (-j0.25) - 2I_x = 0$$

$$V_2 (-j0.75) + j0.25 V_1 - 2I_x = 0 \rightarrow (2)$$

Dependent source Constraint :-

Result :-

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$$I_x = \frac{V_1}{-j2.5}$$

Put I_x in eq (2) :-

So eq (1) becomes

So eq (2) becomes

$$V_2 = -0.86 + j0.85j$$

$$V_2 = -j0.73 V_1 \rightarrow (3)$$

Put V_1 in eq (3)

Put eq (3) in eq (1)

$$V_2 = 0.6 + j0.40$$

c) Write the steady state expression for i_x .

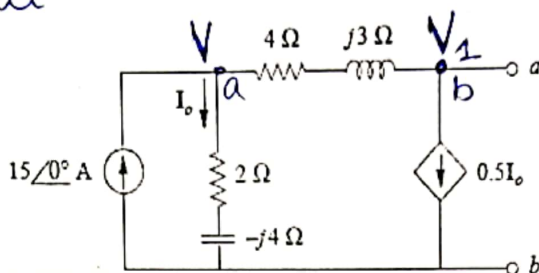
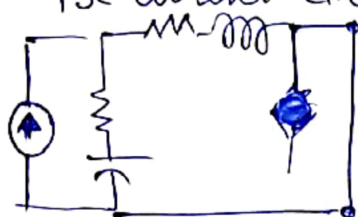
$$i_x = \frac{V_1}{-j2.5}$$

$$I_x = \frac{-0.56 + 0.85j}{-j2.5}$$

$$I_x = -0.34 - 0.224j$$

$$i_x = 0.40 \cos(4t - 33.37^\circ)$$

d) Find the Thevenin Equivalent impedance, Z_{TH} with respect to terminals a and b



Node "a" Node voltage :-

$$-15 + \frac{V}{2-j4} + \frac{V-V_1}{4+j3} = 0$$

$$V \left(\frac{1}{2-j4} + \frac{1}{4+j3} \right) - \frac{V_1}{4+j3} = 15$$

$$V(0.26 + j0.08) - V_1(0.16 - j0.12) = 15 \rightarrow (1)$$

Node "b" equation :-

$$\frac{V_1 - V}{4+j3} + 0.5 I_o = 0$$

$$I_o = \frac{V}{2-j4}$$

$$\frac{V_1 - V}{4+j3} + 0.5 \times \frac{V}{2-j4} = 0$$

Dependent
Source
Constraint

$$I_o = \frac{V}{2-j4}$$

$$\Rightarrow V_1 = (1 - j0.55)V \rightarrow (2)$$

Put eq (2) in eq (1)

$$\Rightarrow \text{So } V = 0.148 - 0.89j \rightarrow (3)$$

Put eq (3) in eq (2)

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So $V_1 = -0.32 - j1.064 \Rightarrow V_{Th} = -0.32 - j1.064$

finding i_{sc} by making a and b terminal
join :- So $i_{sc} = 15(2-j4)/(2-j4 + 4+j3) = 6.48 - j8.91$

for t-sc

$$Z_{Th} = \frac{V_{Th}}{i_{sc}}$$

$$Z_{Th} = \frac{-0.32 - j1.064}{6.48 - j8.91} = 0.06 - j0.08 \Omega$$

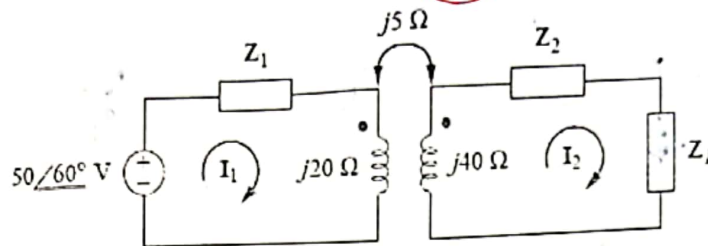
Thevenin equivalent circuit)

Question #2

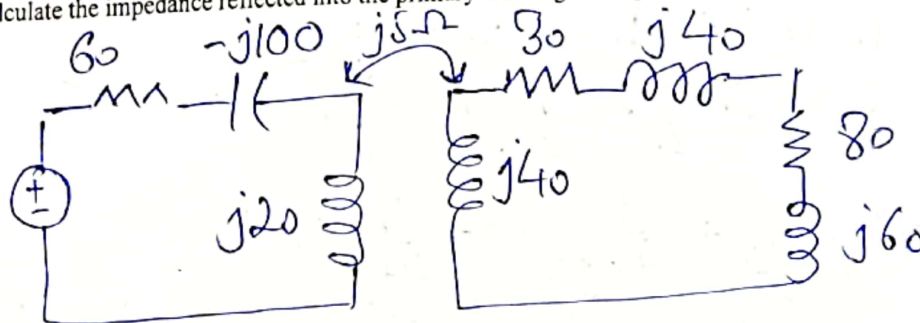
CLO 1

[5 + 15 = 20 marks]

- a) A linear transformer couples a load $Z_L = 80 + j60 \Omega$ to a sinusoidal voltage source. Take $Z_1 = 60 - j100 \Omega$ and $Z_2 = 30 + j40 \Omega$.



Calculate the impedance reflected into the primary winding and input impedance.



Reflected impedance :-

$$Z_{Rl} = \frac{(NM)^2}{|Z_{22}|^2} \times Z_{22}^*$$

$$Z_{Rl} = \frac{(5)^2}{(\sqrt{(110)^2 + (140)^2})^2} \times (110 - j140)$$

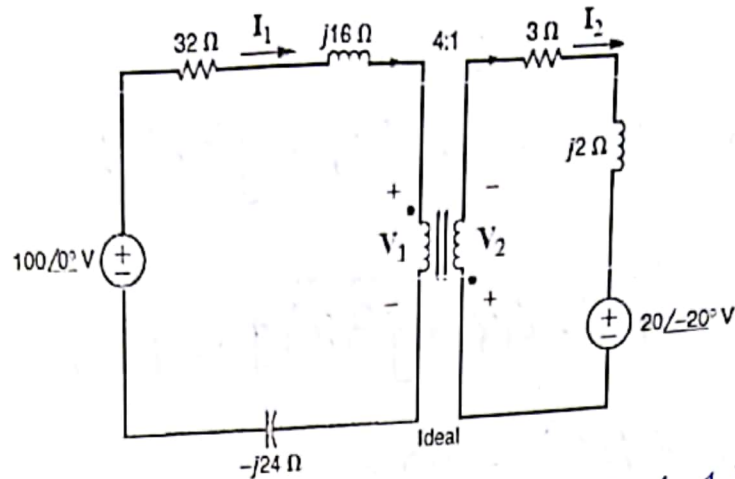
$$Z_{Rl} = 0.086 - j0.1104$$

Input Impedance :-

$$Z_{11} = 60 - j100 + j20 + Z_{Rl}$$

$$Z_{11} = 60 - j80 + Z_{Rl}$$

b) Determine I_1, I_2, V_1 and V_2 in the figure shown below:



Solution:-

$$\text{as } \frac{V_1}{N_1} = + \frac{V_2}{N_2}$$

$$\frac{V_1}{4} = \frac{V_2}{1}$$

$$\Rightarrow V_1 = 4V_2 \rightarrow (1)$$

Similarly:

$$I_1 N_1 = - I_2 N_2$$

$$I_1 \times 4 = - I_2 \times 1$$

$$I_2 = -4I_1 \rightarrow (2)$$

Finding

$$\text{where } V_2 = I_2 \times R$$

$$-V_2 = I_2 (3 + j2) \rightarrow (3)$$

Put Eq (3) in eq (1)

So eq (1) becomes,

$$V_1 = 4 \times (3 + j2) I_2$$

$$\text{where } I_2 = -4I_1$$

So

$$V_1 = 4 \times (3 + j2) \times (-4I_1)$$

$$V_1 = -16(3 + j2) I_1$$

$$V_1 = (-48 + j32) I_1$$

$\rightarrow (4)$

Finding I_1 :-

$$I_1 = \frac{100 - V_1}{32 + j16 - j24}$$

$$I_1 = \frac{100 - V_1}{32 - j8}$$

$$I_1 (32 - j8) = 100 - V_1 \rightarrow (5)$$

Put eq (4) in eq (5);
so eq (5) becomes;

$$I_1(32 - j8) = 100 - (-48 + j32)I_1$$

$$I_1(32 - j8) + (-48 + j32)I_1 = 100$$

$$I_1(-16 + j24) = 100$$

$$I_1 = \frac{100}{-16 + j24}$$

$$I_1 = -1.92 - j2.88 \text{ A}$$

Put I_1 in eq (4)

so;

$$V_1 = (-48 + j32) \times (-1.92 - j2.88)$$

$$V_1 = 184.32 + j76.8 \text{ V}$$

V_2 :-

$$\text{as, } V_1 = 4V_2$$

$$4V_2 = 184.32 + j76.8$$

$$\Rightarrow V_2 = 46.08 + j19.2 \text{ V}$$

I_2 :-

$$\text{as } I_2 = -4I_1$$

$$I_2 = -4(-1.92 - j2.88)$$

$$I_2 = 7.68 + j11.52$$